SINGAPORE DIMENSIONS • KINDERGARTEN (KA) • CHAPTER 3

Comparing Numbers within 10

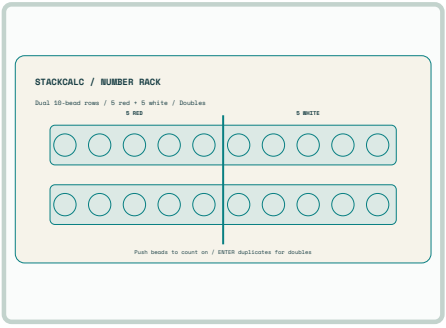
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.CC.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Slide 8 beads left on top wire and 5 beads on bottom wire.

Sketch Model on Paper:
Draw matching one-to-one correspondence lines between two sets.

3. Verify with SC-6 (Abstract)

RPN Stack Register Flow:
Subtracting X from Y reveals 'how many more' in register X.

Self-Check Keystroke Sequence:

8 enter 5 -

Expected LCD: 3 more beads

1 **Compare 7 and 4. How many more is 7?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Compare 9 and 6. What is the difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Compare 10 and 2. How many more is 10?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Compare 6 and 6. What is the difference?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 3

Comparing Numbers within 10 • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Number Rack (2-Wire Stand) & Ten-Frames

CAD Blueprint: number_rack

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White & Signal Red Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Compare 7 and 4. How many more is 7?** Standard: CCSS.MATH.K.CC.C.6	3	[7] [ENTER] [4] [-]
2	**Compare 9 and 6. What is the difference?** Standard: CCSS.MATH.K.CC.C.6	3	[9] [ENTER] [6] [-]
3	**Compare 10 and 2. How many more is 10?** Standard: CCSS.MATH.K.CC.C.6	8	[1][0] [ENTER] [2] [-]
4	**Compare 6 and 6. What is the difference?** Standard: CCSS.MATH.K.CC.C.6	0	[6] [ENTER] [6] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Clarify that subtraction finds difference between sets, not just removal.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.